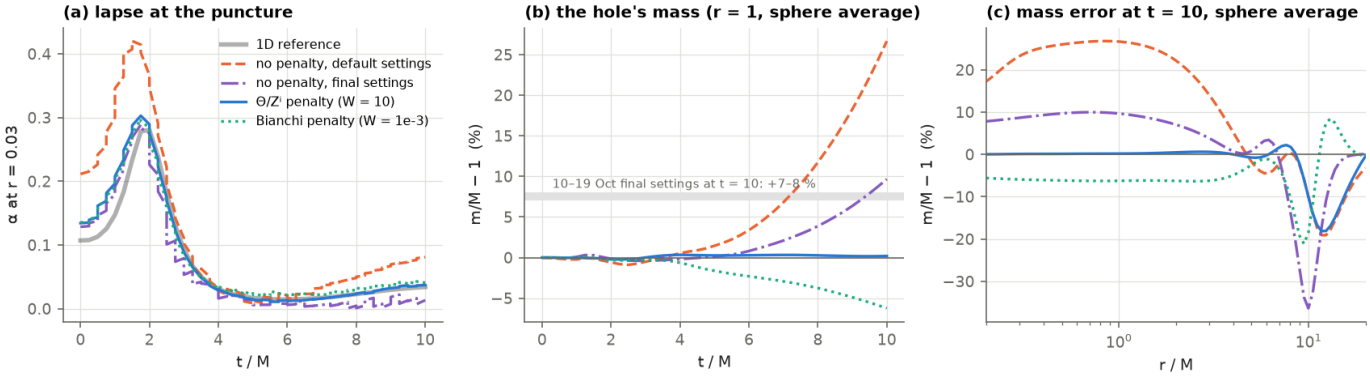
Black-hole puncture CCZ4 PINN: constraints, mass and drift

Work of 20–28 October 2026 · O. Reula, with Claude · same code and problem as the 10–19 October summary (J. Massó): one 6×40 network per 0.25 M window, single puncture m = 0.5, t ≤ 10 M, SSBroyden throughout. All results graded against the collaborators' 1D spherical reference (dr = 0.005) and the Misner–Sharp mass, not the training loss. The control run (final settings, no penalty) was added on 28 October.

Bottom line. The trained networks are not Ricci-flat near the horizon, and the residual loss cannot see why: a constraint-violating solution still satisfies the CCZ4 evolution equations. Adding the CCZ4 constraints Θ and Z^i (zero for a true solution) to the loss fixes the black hole: to t = 10 M its mass stays within **0.35 %** (final settings without it: +9.7 %, as the 10–19 Oct summary's +7–8 %), the lapse at the puncture within 0.003 of the reference, and the lapse error near the hole drops 10×. What the penalty does not change is a mass **deficit** that leaves the hole at t ≈ 2 and travels outward with the gauge pulse, growing to about −20 % (sphere average) at r ≈ 10–12 by t = 10, as with the default settings. The penalty does not remove it, but it matters: with the final settings and no penalty the deficit doubles, to −36 %. Both penalised runs also hold the puncture in place: drift 0.010 and 0.007 by t = 10, against 0.13 without a penalty (defaults) and 0.014 for the final settings alone (0.023 in the 10–19 Oct summary). A second network trained over the whole run on the true Hamiltonian and momentum constraints removes most of the deficit (**−18 % → −5 %** at t = 10), at the price of a lapse error about 2× larger (section 7). Point-symmetric sampling clouds stop it too (25× less).



Overview, t ≤ 10 M. (a) α at r = 0.03 against the reference. (b) Misner–Sharp m/M − 1 at r = 1, outside the horizon (r ≈ 0.25), at each window end; the band is the 10–19 Oct final settings at t = 10. (c) m/M − 1 against r at t = 10. Mass errors are averaged over spheres (48 directions) about the tracked puncture. "No penalty, default settings", "no penalty, final settings", "Θ/Z^i penalty" and "Bianchi penalty" are the runs `single_a`, `long_ctl` (the control run), `long_con10` and `long_bia3`; their settings are in the table of section 1.

1. Runs, settings and collocation points

Two configurations are compared. The **defaults** are the code's default flags, the 10–19 Oct summary's "first baseline", used by the run without a penalty (`single_a`). The **final settings** are that summary's best configuration, used by both penalised runs and by the control run (`long_ctl`). The changes are more points around the hole, twice the QN iterations and, for the penalised runs, the penalty:

per window (0.25 M)	defaults	final settings	distribution
interior, core cloud	–	**48 000**	Gaussian, σ = 0.5, about the tracked puncture
interior, focus cloud	48 000	48 000	Gaussian, σ = 2, about the tracked puncture
interior, bulk	32 000	32 000	uniform in the ball r ≤ 24
initial and end surfaces, each	8000	**12 800**	focus + bulk (60/40), plus 4800 core points
outer boundary (Sommerfeld)	16 000	16 000	on the sphere r = 24
QN iterations	12 000	**24 000**	SSBroyden; all points redrawn every 250
redraws of the points	48	**96**	
constraint penalty	none	**Θ/Z^i, W = 10 or Bianchi, W = 1e-3**	from t = 0 (8 windows to t = 2, then continued to t = 10); none in the control run

Everything else is the same: one 6×40 tanh network per window, m = 0.5, shift-driven puncture tracking, no point within 0.02 of the puncture, interior times from a truncated exponential at the window start widened to uniform by 70 % of the redraws. The defaults run used the QN code before its 19 Oct update (scale invariance, now on by default). The Θ/Z^i penalty uses all 128 000 interior points; the Bianchi term uses the 4096 last ones (core cloud). With the final settings, 29 % of the interior points lie at r < 1, 43 % at 1–5, **10 % at 5–15** (24 % of the volume, where the outgoing mass deficit of section 5 travels) and 19 % at 15–24.

2. The loss is not the accuracy; the reference is

With the final settings the code reproduces the 10–19 Oct numbers against the reference: α error at the puncture **0.022 / 0.032** at t = 0.25 / 1 (theirs 0.019 / 0.024), 5× better than the defaults (0.108 / 0.153). On fresh points never trained on, the loss is 5–10× higher than the value logged in training, and the logged loss *falls* (1e-6 → 2e-7) over a stretch where the mass error of the same run

grows to +28 %. Every result below is therefore judged by the reference and by the mass, which in vacuum spherical symmetry must equal M at every radius and in any slicing (checked exact to 4e-15 on the initial data).

3. Where the mass error comes from

Each window's network is evaluated at its start and end, so the error at any radius splits exactly into the initial-data fit, the growth *inside* windows, and the jumps at the hand-offs between windows. In `single_a` at t = 10 (r = 1): initial −1e-4, inside windows +0.30, hand-offs −0.015. So the error is **built inside the windows**, not inherited at the hand-offs, and its radial density is positive only at r ≲ 0.3, around the horizon; the integral carries it outward.

The cause is direct. We rebuilt the 4D metric and the Z4 vector Z_μ from the network's CCZ4 fields (Alic et al. mapping; exact Schwarzschild passes it to 7e-15) and evaluated the Z4 field equations $E_{\mu\nu} = R_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu$. In window 1, the rms of E is 0.2 at r ∈ [0.3, 0.7), 7e-3 at [0.7, 1.5) and 4e-4 at [3, 6): **the network's spacetime is not Ricci-flat at the ~5 % level around the horizon**, while the CCZ4 residual there is 2.4e-4. The residual loss sees the evolution equations, and these admit solutions with $\Theta, Z^i \neq 0$, which carry wrong mass. Only the initial data rule them out.

4. Constraint penalties

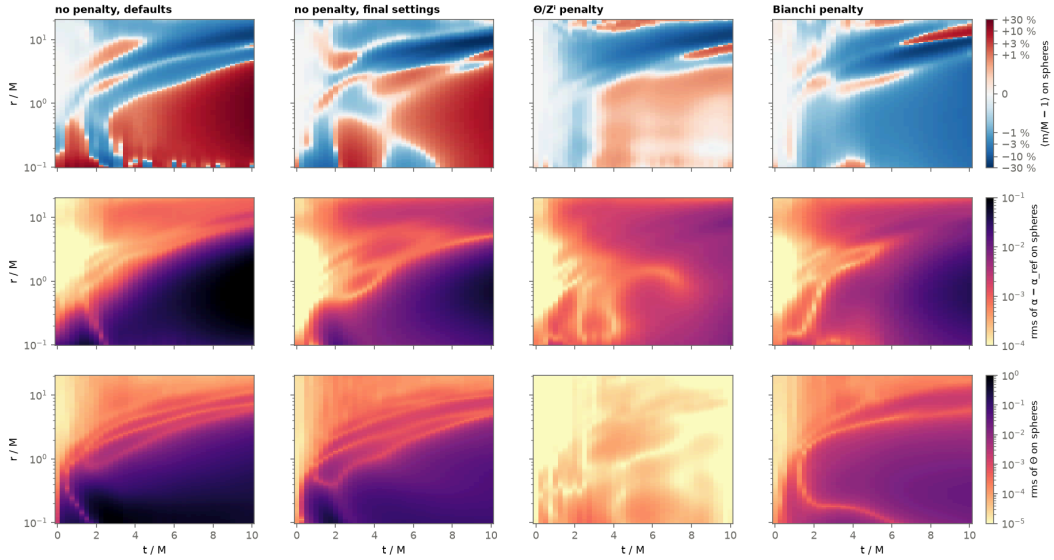
Two ways to rule them out, each a term added to the loss, zero on the true solution:

- **Direct** (`--w_constraint W`): W·mean($\Theta^2 + |Z^i|^2$) over the interior points. Cheap.
- **Contracted Bianchi** (`--w_bianchi W`): for any metric and covector, $\nabla^\mu(E_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}E) = \Box Z_\nu + R_{\nu\mu}Z^\mu$. Z obeys a linear wave equation sourced only by the divergence of the field-equation residual, which is the part of the error that feeds the constraint modes. The term W·mean(B^2), B = □Z + R·Z, on 4096 core points, checked to 5e-16 on random metrics. It needs third derivatives of the network and costs ~35× the residual per point.

At t = 2 (8 windows), both cut the core errors ~10× against the same run without them: core α error 1.2e-2 → 7e-4 (direct) / 1.1e-3 (Bianchi), max Θ 0.45 → 3e-4 / 1.4e-2. Continued to t = 10:

t = 10	no penalty, defaults	no penalty, final (control)	direct, W = 10	Bianchi, W = 1e-3	10–19 Oct final
α at r = 0.03 (reference 0.034)	0.081	0.013	**0.037**	0.041	0.041
α error, core 0.05 ≤ r < 0.5 / mid 0.5 ≤ r < 5	6.6e-2 / 1.0e-1	3.8e-2 / 3.9e-2	**6.0e-3 / 3.5e-3**	2.2e-2 / 2.7e-2	–
max |Θ|	0.42	0.13	**3e-5**	2.9e-2	–
mass of the hole, m/M − 1 at r = 1	+27 %	+9.7 %	**+0.19 %**	−6.2 %	+7–8 %
worst outer deficit (radius)	−19 % (12.4)	−36 % (10.0)	−18 % (11.7)	−21 % (9.5)	≲ 17 % (4–12)
puncture drift $\|x_p\|$	0.13	0.014	0.010	0.007	0.023

The direct term is the clear choice for the hole: Θ falls 10^4×, and the lapse and the mass at the puncture match the reference. The control run shows this is the penalty, not the settings: with the final settings alone the lapse at the puncture falls to 0.013 (reference 0.034; it touches zero at t ≈ 9.75), the core lapse error is 6× the penalised run's, and the hole is 9.7 % too heavy, the 10–19 Oct summary's +7–8 %. The Bianchi term, the more physical idea, does less at the weight tried and makes the hole too light (−6 %).



Errors on spheres, radius against time (columns: the runs of the table above). Top: the Misner–Sharp error averaged over each sphere (red = too heavy, blue = too light). Middle: rms over the sphere of α − α_{ref}. Bottom: rms of Θ. Evaluated at t = 0 and every window end, 48 directions × 100 radii.

5. What is left: an outgoing mass deficit

The maps separate two errors. Inside the outgoing front the hole's mass is what the penalties fix: steadily too heavy without them (+27 %), right with the direct term, uniformly too light with the Bianchi term. The second error is common to all four runs: a deficit band leaves the hole at t ≈ 2 and moves out along the path of the gauge pulse, reaching −18 to −21 % at r ≈ 10–12 by t = 10, and −36 % at r ≈ 10 with the final settings and no penalty. This matches the 10–19 Oct observation (“up to 17 % at r ≈ 4–12 where the

gauge pulses pass"). The lapse error and Θ follow the same outgoing paths. Along a single axis the deficit can be twice its sphere average (–36 % on the ±x rays of the direct-penalty run), so it is not spherical. Where the pulse travels (r ≈ 5–15) every run rests on the sparse uniform sample, since the core and focus clouds sit at the hole, so that region is the first candidate. The control run settles it: the final settings alone do not reduce the deficit; without the penalty it grows twice as fast after t ≈ 5 and reaches –36 %, while Θ grows 20× between t = 4 and 10. The settings and the penalty act on different errors, and neither removes this one.

6. The puncture drift: sampling and constraints

A single puncture should stay at the origin; in the defaults run it reaches 0.13 by t = 10. With puncture tracking off the push at the origin is 10× smaller. Holding the clouds at the origin only halves it, while drawing every cloud in point-reflected pairs about its centre stops it:

t = 4, defaults	**drift**	**|β| at origin, t = 3**	**non-spherical part**
clouds follow the tracked puncture	0.020	3.5e-2	0.6
clouds held at the origin	0.010	1.8e-2	0.32
point-symmetric clouds (`--symmetric_focus`)	**0.0008**	**8.3e-4**	**0.17**

The seed is the random lopsidedness within each cloud, which tracking amplifies. The constraint penalties control the drift almost as well, *without* symmetric clouds:

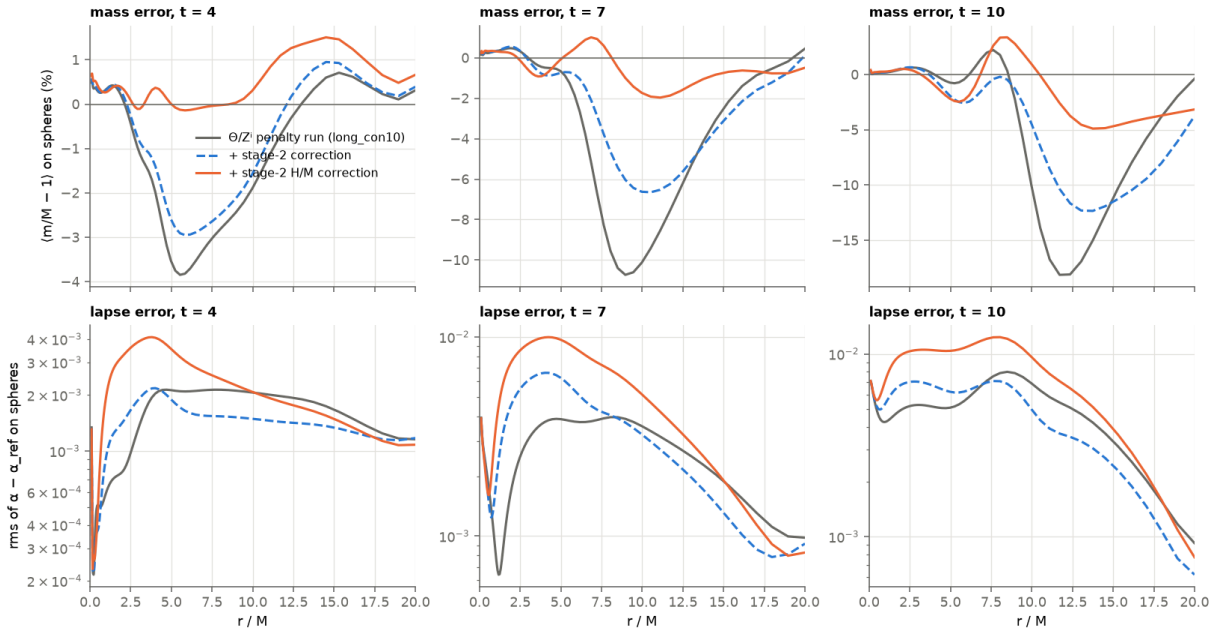
puncture drift $\|x_p\|$	**t = 2**	**t = 3**	**t = 5**	**t = 7.5**	**t = 10**
no penalty, defaults	1.4e-3	1.0e-2	2.7e-2	6.0e-2	0.13
no penalty, final settings (10–19 Oct)	–	–	–	–	0.023
no penalty, final settings (control)	6.0e-4	2.4e-3	5.1e-3	8.3e-3	1.4e-2
Θ/Z^i penalty	3.5e-4	7.7e-4	9.4e-4	5.7e-3	**1.0e-2**
Bianchi penalty	3.9e-4	6.1e-4	1.2e-3	1.7e-3	**6.8e-3**

To t = 5 both stay near 1e-3, the level of the symmetric clouds at t = 4, and at t = 10 they are 13–19× below the defaults and 1.4–2× below the final settings alone (control run; 0.023 in the 10–19 Oct summary). Why, measured on spheres of radius 0.3 about the tracked puncture (`tools/drift_dipole.py`): the tracker moves the puncture with $-\beta(x_p)$, and the Gamma-driver feeds β from the dipole (sphere average) of $\hat{\Gamma}^i$, which CCZ4 splits exactly as $\hat{\Gamma}^i = \tilde{\Gamma}^i[\tilde{\gamma}] + 2Z^i/\chi$. Without a penalty the conformal metric grows a large dipole, $|\langle\tilde{\Gamma}^i\rangle| \approx 0.13$ at t = 10, that $\hat{\Gamma}^i$ mostly does not follow: $2Z^i/\chi \approx -\tilde{\Gamma}^i$, a constraint violation the evolution equations allow. The ~10 % that does reach $\hat{\Gamma}^i$ ($|\langle\hat{\Gamma}^i\rangle| \approx 1.4\text{e-}2$) drives β (3.9e-2) and the drift. With the penalties the metric's dipole is 20–40× smaller (3.5e-3 Θ/Z^i, 6e-3 Bianchi), and $\hat{\Gamma}^i$ and β 5–10× smaller. The asymmetry enters the metric through the constraint-violating mode, and the penalties close it. The final settings alone also drift little (0.014), but their hole and outgoing deficit are worse. Symmetric clouds leave the mass error unchanged; drift and mass are separate problems. The penalised runs drift 10× less without symmetric clouds.

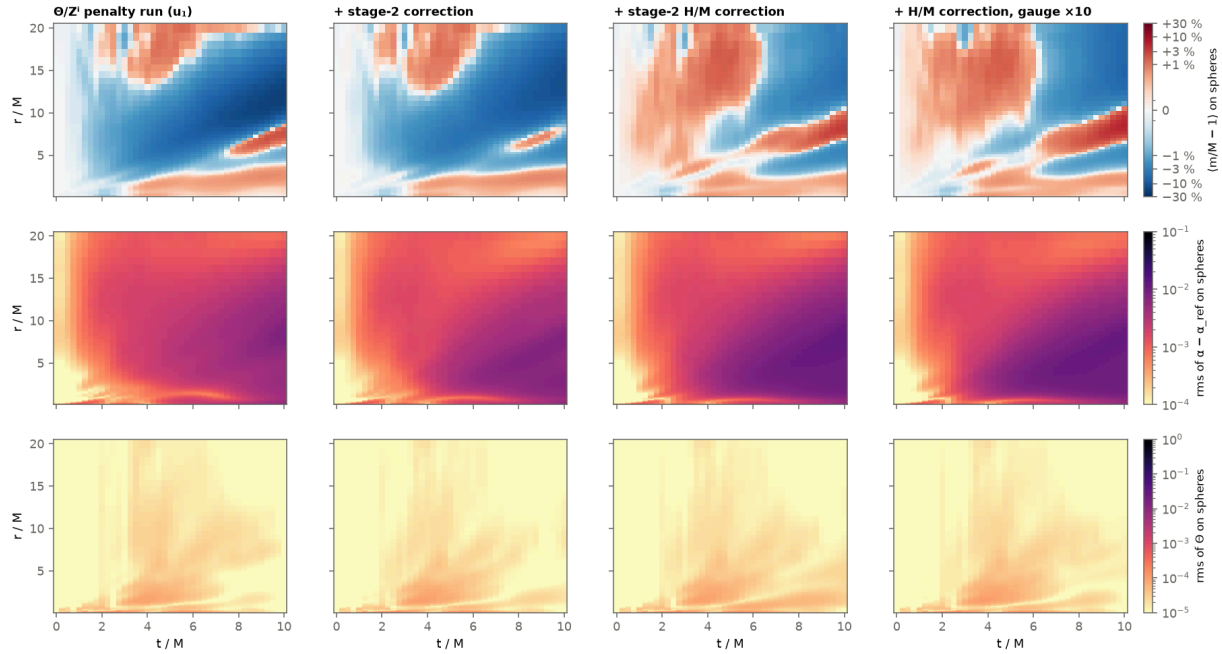
7. Stage 2: one correction network over the whole run

The deficit is a violation of the true ADM constraints, $H = R + K^2 - K_{ij}K^{ij}$ and $M_i = D_jK^j_i - \partial_iK$. Θ and Z^i can vanish while these do not. On spheres, $dm/dR = R^2\langle H\rangle/4$; in the Θ/Z^i run $\langle H\rangle$ integrates to –15 % of the –18 % at r = 11, t = 10, and it is below 1e-3 at t = 0. So it is built during the evolution, inside the windows (–21.5 % at r = 10) rather than at the hand-offs (–0.8 %). To correct the propagation across windows, we froze the finished Θ/Z^i run u_1 and trained a single correction network N over all of $0 \le t \le 10$, so that $u = u_1 + s\cdot N$ with $s = \varepsilon\,(t/T)(1 - r^2/R^2)(1 - e^{-r^2/r_c^2})$. The factor keeps the initial data, the outer boundary and the puncture exactly as in u_1. It is trained on the residual of the sum: N is 4 × 64, with 3 200 points per window span spread evenly in the ball (128 000 per draw, redrawn every 250 steps), 24 000 SSBroyden steps, and ε = 0.01. The run took 1.7–2 h on one A30. Two versions were trained: the same loss as u_1 ("stage-2 correction"), and the same loss plus $\text{mean}(\chi^2(H^2 + |M|^2))$ ("stage-2 H/M correction"). Both are graded as u_1 + correction against the reference.

worst outer deficit (sphere average)	**t = 4**	**t = 6**	**t = 8**	**t = 10**	**α error, r = 3–6, t = 10**
Θ/Z^i penalty run (u_1)	–3.8 %	–8.1 %	–14 %	–18 %	5.2e-3
+ stage-2 correction	–3.0 %	–4.8 %	–8.8 %	–12 %	6.6e-3
+ stage-2 H/M correction	**–0.1 %**	**–0.8 %**	**–2.9 %**	**–4.9 %**	1.1e-2
+ stage-2 H/M correction, gauge rows ×10	–0.4 %	–0.5 %	–2.9 %	–6.7 %	1.0e-2



Stage 2, by radius. Top: sphere-averaged m/M – 1. Bottom: rms over the sphere of α – α_{ref}. The run u_1 (grey) and the two corrections on top of it.



Stage 2, radius against time (linear radius). Rows as in section 4's maps: sphere-averaged mass error, rms lapse error, rms Θ. Columns: the run u_1 and the three corrections of the table. In the H/M columns the deep blue band of the outgoing deficit is gone beyond r ≈ 10 until late times. It leaves a weaker band behind it that is too heavy (up to +3 % at r ≈ 7–9, t = 10), and the lapse error grows over r ≈ 2–8 from t ≈ 4 on.

The H/M term removes most of the outgoing deficit, and the hole stays at +0.2–0.3 % in every version. What it costs is the lapse: about 2× the run's error at r = 3–6. Weighting the gauge equations 10× does not recover it, so this is not a weighting problem. The mass is gauge-invariant, but the lapse is not, and the correction reaches the constraints partly through a different slicing from the reference's. Θ stays small (1–2e-5).

Next steps

- **Windowed H/M run** (running, about half done): the 40-window run with the H/M term in every window, to see whether the deficit can be prevented rather than corrected, and with what effect on the lapse.
- **More interior points where the pulse travels** (r ≈ 5–15), e.g. a shell cloud following the pulse, to test whether the outgoing deficit is under-sampling.
- **Combine** the direct penalty with point-symmetric clouds and take it to the binary, where both the drift and the mass of each hole matter.